

WEAPON

missile submarine

6,000 tonne submarine (not the elderly Charlie class N-sub as thought earlier) will make India the world's sixth nation to operate a "boomer".

Part of the acceleration in the programme has to do with the rapid build-up of Chinese nuclear forces. China operates 10 nuclear submarines, and in the past year, has fielded as many as three new Jin-class SSBNs, each carrying 12 submarine-launched ballistic missiles (SLBM). "Given the growing military asymmetry with China, India's need for a reliable nuclear deterrent that can survive a first strike has never been greater," says strategic expert Brahma Chellaney.

ATV is in line with India's nuclear doctrine enunciated in 1999, which calls for its nuclear forces to be effective, enduring, diverse, flexible and

responsive to the requirements in accordance with the concept of credible minimum deterrence. The doctrine calls for high survivability against surprise attacks and for a rapid punitive response. A nuclear submarine that can remain submerged almost indefinitely and cannot be detected underwater, therefore, meets all these criteria and offers an almost invulnerable launch platform for nuclear weapons. For a country like India with a no-first use policy, it is vital because it prevents a potential adversary from launching a crippling first strike that can knock out all nuclear weapons (see box). It also allows India to inflict considerable damage to the aggressor. "One submarine carries at least 12 missiles with Multiple Independently Targetable Reentry Vehicles, which could mean as many as 96 warheads. When such a submarine goes out to the sea, that many missiles are removed from our own territory. The enemy's targeting of that many sites gets neutralised," says Rear Admiral

